Autonomous Zamboni Overview

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Contents

1 Introduction

This document describes a general overview of the Autonomous Zamboni simulation setup and ALKU-1 autonomous vehicle control system. The purpose of the project is to develop and implement a fully autonomous ice resurfacing machine.

2 Software tools

The software development is done in Visual Studio Code. The development is done with Ubuntu 24.04 Operating System. ROS2 (Jazzy) is used for building the application. The URDF model has been developed with the xacro specification. It also features a special Zamboni body model that has been imported as a .stl file from Thingiverse and integrated into the model. The physics are simulated in Gazebo Harmonic. Additional vizualisation is observed with RViz.

3 URDF model of the vehicle

The URDF model of the vehicle has been constructed with the help of xacro properties and macros. The URDF model is described in the ice_resurfacer2.urdf.xacro file. It describes a base link, wheel links, steering links, conditioner link, IMU link, and LiDAR link. The dimensions of the vehicle are described with xacro properties.

4 Ice rink simulation world

The simulation has been developed in Tinkercad. The rink was formed from basic elements, such as cylinders and cubes. Two ice hockey goals were imported from a custom model to serve as props. .STL world models are stored in meshes folder. The Worlds folder includes a Gazebo world file (.sdf), which refers to the .stl mesh to acquire the visual and collision link.

5 Project structure

The project has been modularly divided to different sub-folders. ice_resurfacer_description, ice_resurfacer_gazebo, and ice_resurfacer_nav

6 System architecture

7 Path Planning

8 Localization

9 SLAM